

Practice problems -Session2

Description

a) A retail company wants to segment its customers based on the following categorical features:

- Region (North, South, East, West)
- Preferred Payment Method (Cash, Card, UPI, Wallet)
- Shopping Frequency (Low, Medium, High)
- Loyalty Membership (Yes, No).

Can you apply the K-Means algorithm for the above dataset and if it is suitable, illustrate the steps. Otherwise, apply an appropriate algorithm and illustrate the steps in detail [5 Marks]

b) Design a simple multilayer perceptron (MLP) to predict student academic performance based on two normalized input features: study hours ($x_1 = 0.6$) and previous exam score ($x_2 = 0.4$). The network should have one hidden layer with a single neuron and a single output neuron, using sigmoid activation functions throughout. Given initial weights $w_1 = 0.3$ (input₁ to hidden), $w_2 = 0.2$ (input₂ to hidden), and $v = 0.4$ (hidden to output), along with biases $b_h = 0.1$ (hidden layer) and $b_o = 0.05$ (output layer), with a learning rate of $\eta = 0.5$ and target output $y = 1$ (indicating pass).

i) Perform one forward pass and compute the predicted output $\hat{y}$ [3 marks]

ii) Calculate the mean squared error loss between the predicted and target values [. 2 marks]

iii) Apply backpropagation and update all parameters using gradient descent by illustrating the steps [5 marks]

Consider a two-class classification problem with a dataset that is perfectly separable by a linear boundary. You are given a choice between a Hard-Margin SVM and a Soft-Margin SVM with a very high value of the penalty parameter C .

a) How would the decision boundaries and margins of these two models compare? [4 Marks]

b) If you add a single, highly influential outlier deep within the opposite class's data, how would each model's decision boundary likely change? Justify your answer. [4 Marks]

c) Based on your analysis, explain why the Soft-Margin SVM is generally a more robust and practical choice for real-world engineering problems. [3 Marks]

d) Discuss the impact on the margin in the following two cases [4 Marks]

i) Removing the support vectors from the data

ii) Removing the non-support vectors from the data

A wildlife researcher is tracking animal sightings in a forest. They have recorded the GPS coordinates (latitude and longitude) of 8 sightings of a rare species. The researcher wants to group sightings that are close together in space, where a group should have at least 2 points within a distance of 2 units from each other. Points that don't belong to any group should be treated as isolated.

Sighting	X	Y
1	1	2
2	2	2
3	2	3
4	8	7
5	8	8
6	25	80
7	24	79
8	25	81

a) Using the appropriate algorithm, identify the clusters formed and any noise points. [7 Marks]

b) Discuss why some sightings are grouped together and why others are classified as noise. [3 Marks]

Consider a Kohonen self-organizing network that can organize/group the data into Five clusters. Consider a two dimensional input vector (x_1, x_2) . Let the weights from first and second input nodes to all the five output nodes be $[0.3, 0.2, 0.1, 0.8, 0.4]$ and $[0.5, 0.6, 0.7, 0.9, 0.2]$, respectively.

a) Find the cluster unit close (BMU) to the data point $(0.2, 0.4)$ using a learning rate of 0.2. What are the updated weights ?

b) Now consider another input data $(0.6, 0.6)$. With a learning rate of 0.1, find the winning cluster for this input. What are the updated weights ? Use the square of the Euclidean distance to find the cluster distances.